# AI Prompting Lab Reflection Report

## Introduction

This AI Prompting Lab helped me understand that the quality of an AI's response depends heavily on the quality of the prompt. Through thirteen practical exercises, I learned how to provide context, evaluate AI outputs critically, use different prompting techniques, and choose the right approach for different tasks. The lab was highly practical because every concept was applied through real examples rather than only theoretical explanations.

## What I Learned From This Lab

By completing this lab, I learned that AI performs best when it is given clear instructions, relevant context, and specific goals. I learned the difference between generic prompts and detailed prompts, and how adding constraints, preferences, and desired output formats significantly improves results.

I also learned:

- How context directly affects the quality of AI responses.
- The difference between information AI already knows and information that requires web search or research.
- How to ask AI to think more deeply about complex decisions.
- How to avoid biased or leading prompts.
- How to use iterative prompting to improve results step by step.
- How AI can analyze images, handwritten notes, and visual content.
- How to generate working applications using structured prompts.
- How to verify calculations by asking AI to write and run code.
- How to compare responses from different AI models to identify strengths and weaknesses.
- How to evaluate AI outputs critically instead of accepting the first answer.

## How My Prompting Improved During The Process

At the beginning of the lab, my prompts were generally short and broad. I would ask a question and expect AI to figure out the missing details.

As I progressed through the exercises, I learned to provide more context, define constraints, specify the audience, and clearly describe the desired output format. Instead of asking simple questions, I started giving AI the information it needed to produce accurate and useful answers.

My prompts became:

- More detailed and structured.
- More goal-oriented.
- Better at defining requirements and constraints.
- More focused on obtaining specific outcomes.
- More effective at reducing ambiguity.

As a result, the quality, relevance, and usefulness of AI responses improved significantly.

## Prompting Techniques That Helped Me The Most

Several prompting techniques proved especially valuable throughout the lab:

### 1. Providing Context Up Front

This was the most useful technique. I learned that giving AI background information, constraints, preferences, and objectives leads to far better responses than asking generic questions.

### 2. Goal – Input – Output Framework

The structured format used for building applications helped me clearly communicate requirements. It reduced confusion and made it easier for AI to generate exactly what I wanted.

### 3. Think Hard Prompting

Adding instructions such as "Think hard before answering" encouraged AI to analyze trade-offs and provide deeper reasoning instead of superficial suggestions.

### 4. Brainstorm → Iterate → Refine

Rather than accepting the first answer, I learned to generate multiple options, provide feedback, and refine responses through several iterations. This consistently produced higher-quality results.

### 5. Cross-Checking Between Models

Comparing responses from different AI systems helped identify blind spots and weaknesses that a single model might miss.

## Challenges I Faced And How I Solved Them

### Challenge 1: Getting Generic Responses

Initially, many responses were too broad and lacked personalization.

**Solution:** I learned to provide detailed context, constraints, and clear instructions before asking for a response.

### Challenge 2: Determining Whether AI Was Accurate

It was sometimes difficult to know whether AI was providing verified information or simply generating a confident answer.

**Solution:** I learned to request citations, ask AI to acknowledge uncertainty, and use web search when dealing with current information.

### Challenge 3: Evaluating Complex Decisions

Some tasks required weighing multiple factors rather than receiving a simple answer.

**Solution:** Using "Think Hard" prompts encouraged deeper reasoning and more balanced analysis.

### Challenge 4: Trusting Calculations

I discovered that AI can sometimes perform calculations through reasoning rather than actual computation.

**Solution:** I learned to ask AI to write and run code, then verify the results using the generated code.

### Challenge 5: Accepting The First Response

At first, I tended to accept the initial answer.

**Solution:** The Brainstorm–Iterate Loop taught me to continuously refine prompts and improve outputs through multiple rounds of feedback.

## Conclusion

This lab significantly improved my understanding of effective AI prompting. The most important lesson I learned is that AI produces the best results when it receives the right context and clear instructions. I developed stronger prompting skills, learned how to evaluate AI responses critically, and gained practical experience using advanced techniques such as contextual prompting, iterative refinement, structured prompting, model comparison, and code verification. These skills will help me use AI more effectively, efficiently, and responsibly in future personal, educational, and professional tasks.